

Article

Research on a Potato Leaf Disease Diagnosis System Based on Deep Learning

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Abstract: Potato is the fourth largest food crop in the world. Disease is an important factor restricting potato yield. Disease detection based on deep learning has strong advantages in network structure, training speed, detection accuracy, and other aspects. This article took potato leaf diseases (early blight and viral disease) as the research objects, collected disease images to construct a disease dataset, and expanded the dataset through data augmentation methods to improve the quantity and diversity of the dataset. Four classic deep learning networks (VGG16, MobilenetV1, Resnet50, and Vit) were used to train the dataset, and the VGG16 network had the highest accuracy of 97.26%; VGG16 was chosen as the basic research network. A new, improved algorithm, VGG16S, was proposed to solve the problem of large network parameters by using three improvement methods: changing the network structure of the VGG16 network from “convolutional layer + flattening layer + fully connected layer” to “convolutional layer + global average pooling”, integrating CBAM attention mechanism, and introducing Leaky ReLU activation function for learning and training. The improved VGG16S network has a parameter size of 15 M (1/10 of VGG16), and the recognition accuracy of the test set is 97.87%. This article used response surface analysis to optimize hyperparameters, and the test results indicated that VGG16S, after hyperparameter tuning, had further improved its diagnostic performance. At last, this article completed ablation experiments and public dataset testing. The research results will provide a theoretical basis for the timely adoption of corresponding prevention and control measures, improving the yield and quality of potatoes and increasing economic benefits.

Keywords: potato; data augmentation; VGG16; network lightweighting; hyperparameter optimization



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1. Introduction

Potatoes have the characteristics of strong environmental adaptability, wide planting area, and abundant nutrients [1]. It is the fourth largest food crop in the world, with a yield second only to corn, wheat, and rice. Due to its short growth cycle and rich nutritional content, it is widely planted in China and has made significant contributions to increasing grain production and promoting farmers' income [2,3].

Disease is one of the important factors affecting potato yield [4,5]. The main potato diseases include early blight, late blight, and viral diseases. The infection process of potato diseases is divided into the incubation period and the symptom period. During the incubation period, there are no obvious symptoms in potato plants, making it difficult to detect diseases. During the symptomatic period, pathogens infect leaves, causing leaves to become mottled and discolored, which affects the normal growth of plants. Accurate identification of diseases in the early stages of potato leaf symptoms is of great significance. On the one hand, it can prevent further damage to potato tubers, and on the other hand, it can prevent the spread of diseases between plants, avoiding a greater impact on potato yield. The annual occurrence area of potato diseases in China is about 4 million hm^2 , with early and late diseases occurring the most frequently and causing the most severe damage. The annual incidence area is 2.95 million hm^2 , accounting for 3/4 of the disease occurrence area [6].

At present, the diagnosis of diseases mainly relies on detecting the environmental or biological causes that cause the disease, as well as identifying the symptoms of crop diseases, which mainly include disease diagnosis methods based on spectral data and disease diagnosis methods based on image data.

(1) Disease diagnosis method based on environmental cause detection.

Karim Foughali et al. [7] deployed a sensor network to collect temperature and humidity information on potato growth environments and uploaded the collected information to a cloud IoT platform for data storage and analysis. Finally, they constructed a decision support platform for potato late blight assessment and prevention. Dai Guowei et al. [8] proposed a potato early and late disease detection model based on machine learning algorithms, which utilizes the FastK means algorithm to detect potato leaf diseases under different weather conditions and time periods. Marwa Radwan et al. [9], in order to determine the correlation with the incidence of potato diseases, studied some key factors, including temperature, humidity, wind speed, and atmospheric pressure. Through complex analysis methods such as K-means clustering, PCA, and copula analysis, these types of data relationships were comprehensively identified. Gao Ronghua et al. [10] took cucumber diseases as the research objects, collected cucumber growth environment information and cucumber disease leaf image information through the Internet of Things, and combined disease text data to achieve multi-structured data learning of “environment + image + disease text description information”, constructed the optimal combination model of disease features, and the model recognition accuracy was 79.4–93.6%.

(2) Disease diagnosis method based on biological genetic detection.

Ye et al. [11] studied barley and barley yellow dwarf virus and observed the infection process of barley yellow dwarf virus on plant chloroplast cells using electron microscopy technology. Paul Rajesh et al. [12] applied disposable polymer microneedles (MNs) on tomato leaves to extract pathogenic DNA from plants that can be used for amplification experiments within one minute and used it for polymerase chain reaction (PCR) to detect tomato late blight. This method helps to quickly prepare samples for molecular diagnosis of various plant diseases in the field.

(3) Disease diagnosis method based on spectral data.

Lin Xinting et al. [13] studied nine spectral preprocessing methods and combined them with experimental results. The preprocessing methods were arranged and improved into 16 spectral preprocessing methods and combined with a continuous projection algorithm, competitive adaptive reweighting algorithm, and genetic algorithm to obtain 64 spectral processing methods for optimizing the original spectral data. To further quantitatively

analyze the different stages of potato early blight infection, the spectral data processed by spectral processing methods will be quantitatively analyzed using a constructed CNN quantitative estimation model. The results indicate that the proposed CNN can effectively classify, detect, and quantitatively analyze potato early blight at different stages of infection. Zhang Xiaoyan et al. [14] innovatively attempted to use near-infrared spectroscopy combined with the SIMCA mode method to distinguish potato ring rot disease from healthy potatoes. The research results indicate that SIMCA pattern recognition based on principal component analysis can effectively distinguish potato ring rot samples, and the recognition and rejection rates of ring rot and healthy potatoes in the model calibration set are both 100%. The recognition rate and rejection rate of ring rot disease in the model validation set were 99.00% and 100%, respectively, while the recognition rate and rejection rate of healthy potatoes were 94.12% and 100%, respectively. The accuracy of the established model is relatively high. Zhuo Wei [15] used potato leaf chlorophyll as a starting point to measure the chlorophyll content in potato leaves and analyze spectral data, obtaining a correlation model between spectral features and chlorophyll content. This model can draw a map of chlorophyll content information in potato leaves, and by visualizing the chlorophyll “holes” in the image, it can help people judge the health status of potato leaves.

(4) Disease diagnosis method based on image data.

Alberta Odamea Anim-Ayeko et al. [16] proposed a ResNet-9 model for detecting wilting symptoms in potato leaf images. After expanding the training set and rigorous hyperparameter optimization, the model is trained using these hyperparameter values, and images are examined on a test set containing 1331. The testing accuracy was 99.25%. Li Na et al. [17] adopted the OTSU-SFLA image segmentation algorithm, which combines the Otsu method (OTSU) with the hybrid frog leaping algorithm (SFLA), to segment images of potato leaf lesions. This algorithm can optimize the segmentation of five common diseases, including potato early blight, and extract and recognize image features by establishing a convolutional neural network recognition model. The experimental results show that the five common diseases of potatoes are recognized. The average recognition rate of diseases is 98.2%. Yang Sen et al. [18] proposed a method for identifying potato diseased leaves based on a combination of deep convolutional neural networks and composite feature dictionaries. The experimental results showed that when the number of visual words in the composite dictionary was 50, the robustness and real-time performance of disease recognition was better; the average recognition rate was 90.83%, as well as the single frame image average time consumed was 1.68 s. Rabbia Mahum et al. [19] proposed an improved deep learning algorithm that classifies potato leaves into five categories based on their visual features: Potato late blight (PLB), potato early blight (PEB), potato roll (PLR), potato wilt (PVW), and potato health (PH). And a pretrained and efficient DenseNet model effectively classified potato leaf diseases using the additional transition layer in DenseNet-201. The algorithm’s performance was evaluated on testing equipment, and an accuracy of 97.2% was obtained.

Environmental factors have gradually become an auxiliary criterion for multi-source data fusion diagnosis of diseases. The disease diagnosis technology based on biological pathogen detection is relatively specialized and complex, but it has limitations. Spectral data are susceptible to interference from factors such as noise, data acquisition environment, baseline drift, etc. They also have the characteristics of numerous bands, large data volumes, complex spectral bands, and different signal-to-noise ratios in different regions. Image technology is susceptible to factors such as lighting and weather.

Disease detection based on deep learning algorithms has the advantages of high transferability and strong universality, as well as strong advantages in network structure, training speed, and detection accuracy for crop disease detection. Therefore, based on deep

learning algorithms and image processing techniques, this article takes common diseases of potato leaves as the research objects and constructs a diagnosis system for common diseases of potato leaves. This study provides accurate identification and diagnostic methods, providing a theoretical basis for scientific and effective disease prevention and control, reducing pesticide misuse and abuse, and achieving green prevention and control.

2. Materials and Methods

2.1. Data Acquisition and Processing

2.1.1. Data Acquisition

The data collection site is located in the potato planting fields in Wuchuan County, Hohhot City, Inner Mongolia Autonomous Region, with the geographical coordinates of 111°27'17" E and 41°6'25" N. The data were collected on 21 August 2024, during the potato growing period. A total of 325 normal potato leaves, 582 potato early blight leaves, and 134 potato virus disease leaves were collected, as shown in Figure 1.

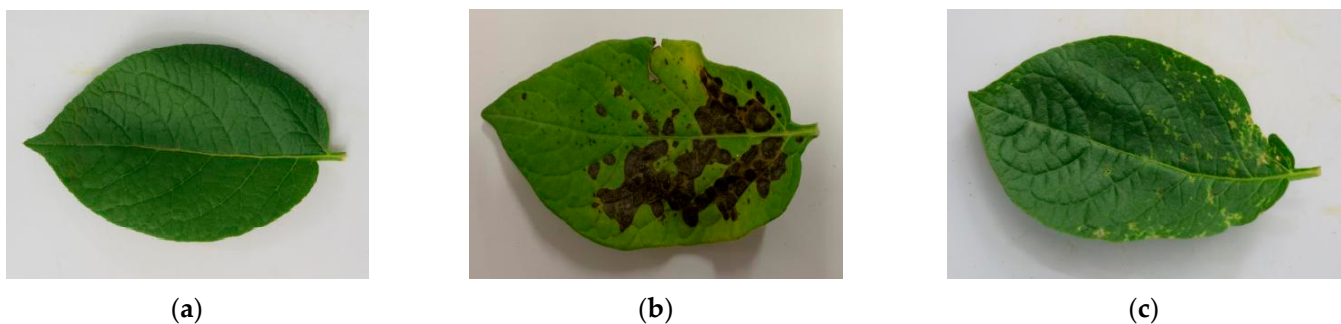


Figure 1. Images of potato normal leaf (a), potato early blight leaf (b), and potato virus disease leaf (c).

2.1.2. Data Augmentation

The method of data augmentation can increase the number of training samples, simulate different interference environments, make the sample dataset diverse (noisy data), and improve the robustness of the training network [20,21]. And in the case of limited training data, data augmentation can effectively alleviate overfitting. By expanding the amount of data, the model is exposed to a more diverse range of inputs during training, avoiding overfitting the training data and enhancing the model's generalization ability [22].

For image data, data augmentation mainly includes methods such as rotation transformation, flipping transformation, scaling transformation, translation transformation, scale transformation, contrast transformation, noise disturbance, and color transformation. Considering the practical application scenarios of diagnostic networks, this article used four methods: rotation transformation, flipping transformation, contrast transformation, and noise disturbance to randomly enhance the dataset.

Rotation: The mathematical transformation is shown in Equation (1). The transformation effect of rotating the disease image by 145° is shown in Figure 2b.

$$\begin{cases} i' = i \cos \theta - j \sin \theta \\ j' = i \sin \theta + j \cos \theta \end{cases} \quad (1)$$

where i' and j' are the rotated coordinates, i and j are the original coordinates, respectively, and θ is the rotation angle.



Figure 2. Example of a rotation transformation: (a) for the image of untreated disease, (b) for the image after 145° of rotation.

Flipping: Taking horizontal mirroring as an example, the image width is f_w , and the pixels (x, y) are swapped left and right around the vertical axis as the center, which can be described by Equation (2). The effect of flipping the diseased leaf image horizontally and vertically is shown in Figure 3.

$$\begin{bmatrix} \tilde{x} \\ \tilde{y} \\ 1 \end{bmatrix} = M_{AF} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, M_{AF} = \begin{bmatrix} -1 & 0 & f_w \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

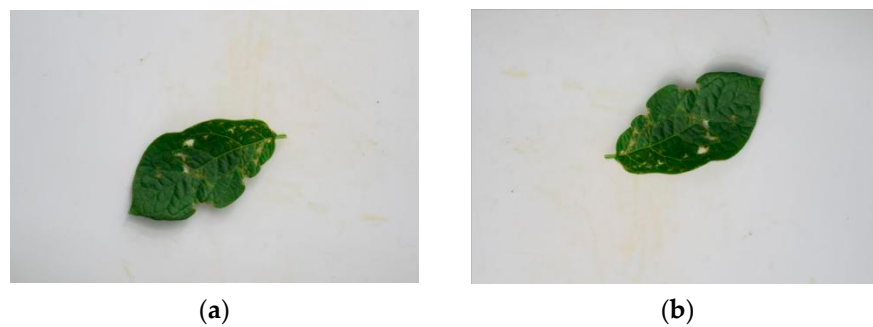


Figure 3. Example of a flipping transformation: (a) for the image of untreated disease, (b) for the image after horizontal and vertical flip.

Contrast: Assuming that the grayscale range $[m, M]$ of the original image $f(x, y)$ is linearly transformed into a new grayscale range $[n, N]$, the transformation formula is shown in Equation (3):

$$g(x, y) = \frac{N - n}{M - m} [f(x, y) - m] + n \quad (3)$$

The effect of insufficient and excessive lighting in the scene simulated by contrast transformation is shown in Figure 4.

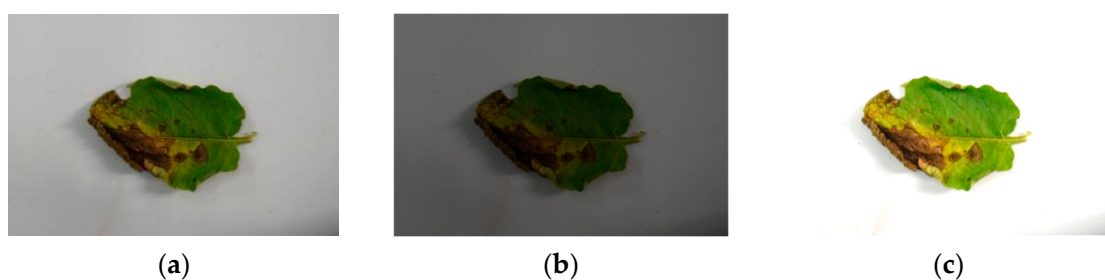


Figure 4. Contrast transformation effect: (a) for the image of untreated disease, (b) for the image of low-contrast transformation, (c) for the image of high-contrast transformation.

Noise: This article added a certain amount of noise to data augmentation for simulation, enhancing the diagnostic network's ability to process noisy images. The effect of adding Gaussian noise with a mean of 0 and a variance of 0.01 to images of diseased leaves is shown in Figure 5.



Figure 5. Noise transformation effect: (a) for the image of untreated disease, (b) for the image after adding Gaussian noise.

2.2. Dataset Construction

After data augmentation, a dataset of common potato leaf diseases with a total of 3129 images was formed, including 1082 images of potato early blight, 1072 images of potato viral disease, and 975 images of normal potato leaf. We randomly divided the disease dataset into a training set and a testing set in a 7:3 ratio. During the training process, the training set will randomly allocate 30% of the images as the validation set to verify the accuracy of the diagnostic network in each training round (epoch). The composition of the dataset is shown in Table 1.

Table 1. Composition of the common potato diseases dataset.

	Potato Normal Leaf (pcs)	Potato Early Blight Leaf (pcs)	Potato Virus Disease Leaf (pcs)	Total (pcs)
Training set (pcs)	683	757	750	2190
Testing set (pcs)	292	325	322	939
Total (pcs)	975	1082	1072	3129

2.3. Basic Network of Potato Leaf Diseases Diagnosis

Convolutional Neural Network (CNN) is a type of feedforward neural network that is particularly suitable for processing data with grid-like topological structures, such as images, sounds, and text. Due to its end-to-end processing approach, which eliminates the need for image preprocessing, the process of its detection task has become very flexible and convenient. With the development of convolutional neural networks, researchers have found that increasing the number of layers can improve the accuracy of the network, but at the same time, it can also make the number of network parameters become large and the calculation speed slow. Moreover, the number of convolutional layers cannot be infinitely increased through simple stacking. After exceeding a certain number of layers, the recognition accuracy of the network will no longer improve, which has become a technical bottleneck for deep learning algorithms.

This article uses the CNN algorithm for image classification. Currently, common image classification networks include VGG16, MobilenetV1, Resnet50, and Vit, which networks have different characteristics. This article adopted these four networks for comparative tests, aiming to select the network with the highest disease recognition accuracy as the basic network for diagnosing potato leaf diseases.

2.3.1. Convolutional Neural Network

VGG16

VGG16 has a simple structure, excellent performance, and generalization ability and is characterized by the use of small convolution kernels and deep network layers. The network structure of VGG16 is shown in Figure 6 [23].

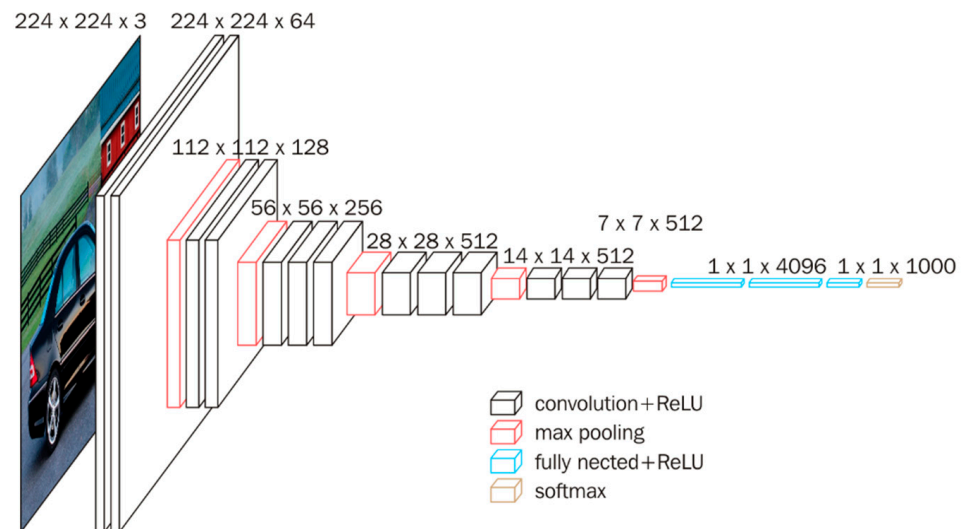


Figure 6. VGG16 network structure.

ResNet50

Deepening the number of network layers can lead to the problem of gradient vanishing, resulting in a decrease in network recognition accuracy. ResNet50 adopts a residual network structure to overcome the gradient vanishing problem and effectively improve the accuracy of the network. The residual network structure of ResNet50 is shown in Figure 7 [24].

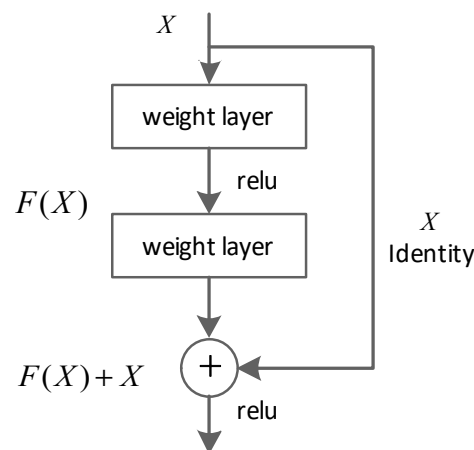


Figure 7. Residual network structure.

Mobilenet V1

MobilenetV1 uses depthwise separable convolution as the basic processing unit for image feature extraction and achieves high accuracy and fewer network parameters through layer stacking. The processing structure diagram of depthwise separable convolution is shown in Figure 8 [25].

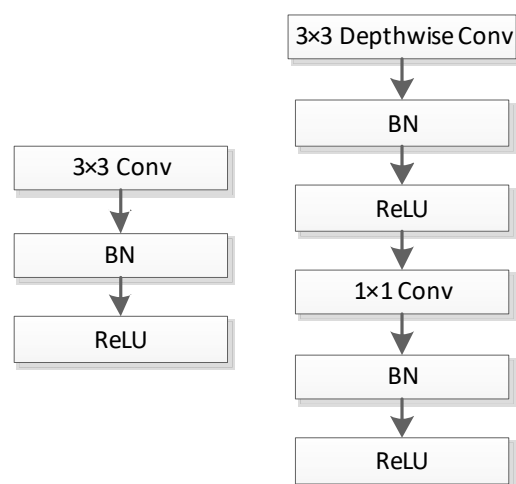


Figure 8. Depthwise separable convolutional structure diagram.

Vit

Vit (Vision Transformer) is an image classification network based on a Transformer structure, which is a graph neural network. The process of Vit processing images is shown in Figure 9 [26].

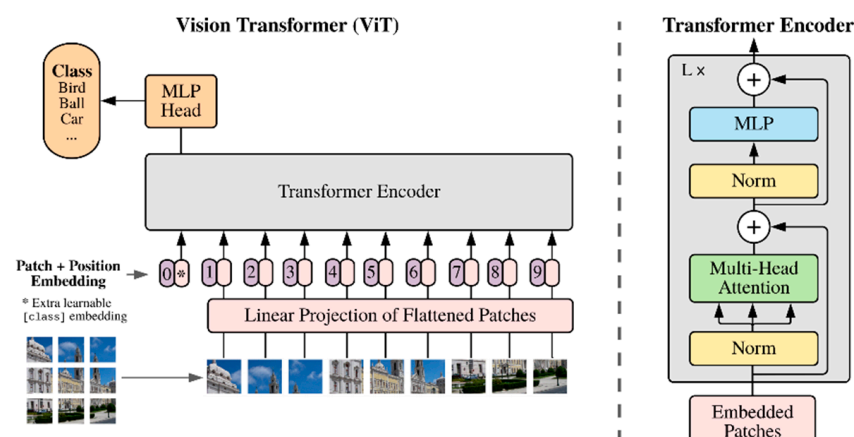


Figure 9. Processing of images by the Vit network.

2.3.2. Network Training and Evaluation

Train the potato leaf disease dataset using four image classification networks: VGG16, MobilenetV1, Resnet50, and Vit. Use the validation set to verify the recognition accuracy of the trained network and select the network with the highest accuracy. The training environment for each classification network is shown in Table 2.

Table 2. Network training environment.

Configuration	Version
Processor	AMD Ryzen 7 4800H 32G
GPU	NVIDIA GeForce GTX 1650 4G
Operating System	Windows 10
Compiler	PyCharm Community Edition 2020.3.2 x64
Virtual Environment	TensorFlow-gpu 1.13.1
	Keras = 2.1.5
	Numpy = 1.17.4

The hyperparameters of the network are set based on empirical values, with 100 iterations, a batch size of 32, and a learning rate of 0.001. The training adopts a learning strategy of learning rate exponential descent and early stopping. After training, the diagnostic results of each image classification network on the potato leaf disease dataset are shown in Figure 10. The validation set recognition accuracies of VGG16, MobilenetV1, Resnet50, and Vit networks were 97.26%, 79.76%, 94.52%, and 95.89%, respectively. The test set recognition accuracies were 97.23%, 59.96%, 81.36%, and 97.02%, respectively. The variation curves of validation set recognition accuracy and loss with increasing epochs are shown in Figures 11 and 12.

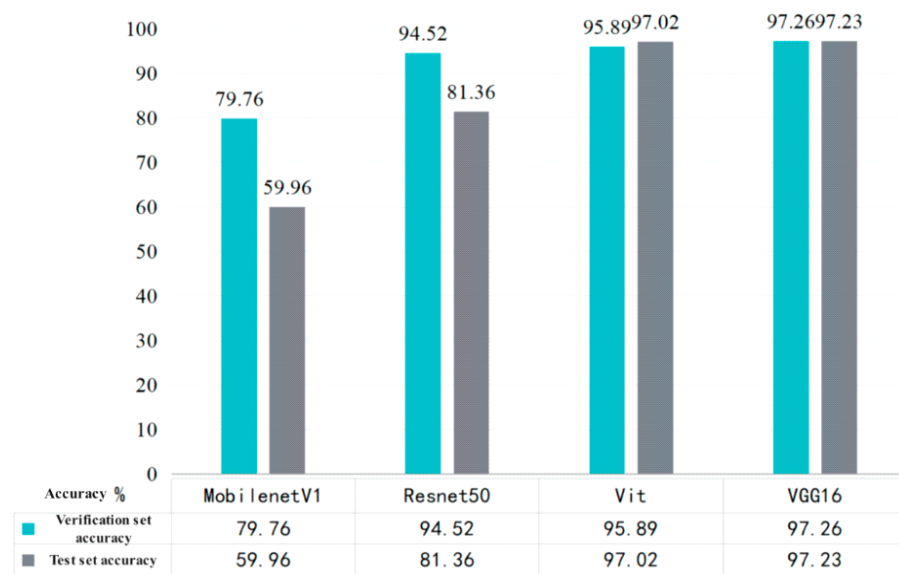


Figure 10. Results of each image classification network for the potato leaf disease diagnosis task.

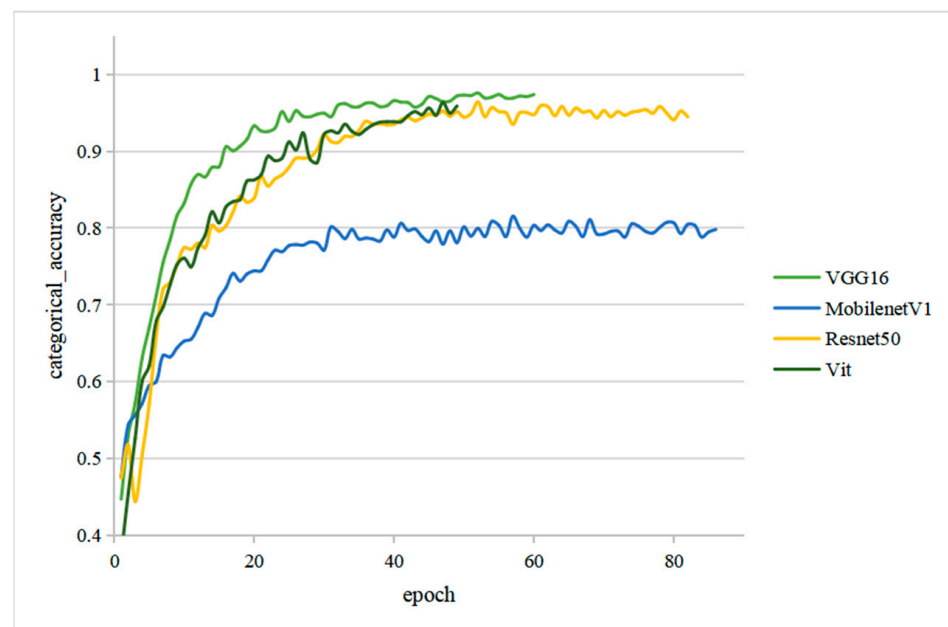


Figure 11. Variation curve of validation set recognition accuracy with epoch for each image classification network.

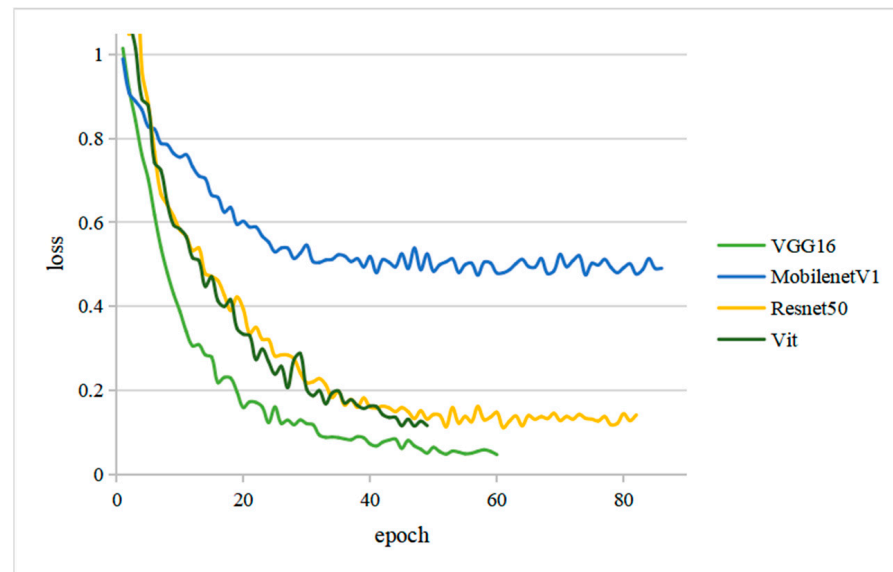


Figure 12. Variation curve of loss with epoch for each image classification network.

The training results show that the four networks used in this article have demonstrated good convergence characteristics for potato leaf disease diagnosis, with VGG16 performing the best. Both in terms of convergence speed and convergence level, VGG16 is significantly better than the other three classification networks, indicating that the VGG16 network structure has a higher completion rate for the research task in this paper. This article will use VGG16 as the basic network for further research on potato diagnosis tasks.

2.4. Network Optimization Strategy

2.4.1. Lightweight Improvement

The network VGG16 adopts a network structure of “convolutional layer + flattening layer + fully connected layer”, which is the main reason for a large number of network parameters, accounting for 89.36% of the total number of network parameters. Therefore, this article uses the Global Average Pooling process (GAP) idea to improve the network structure of the “convolutional layer + flattening layer + fully connected layer” to the “convolutional layer + Global Average Pooling (GAP)” structure. GAP [27] adopts a process of channel averaging, which can greatly reduce network parameters and avoid overfitting during training. In addition, GAP can increase the interpretability of the network’s recognition mechanism. The Global Average Pooling process (GAP) is shown in Figure 13.

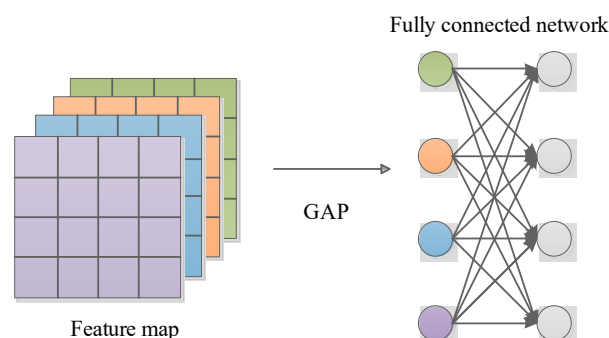


Figure 13. Global Averaging Pooling process.

2.4.2. Attention Mechanism Introduction

The channel attention mechanism introduces a weight evaluation mechanism for each channel and uses adaptive learning methods to distinguish the importance of each channel, achieving directional allocation of network computing power. The spatial attention mechanism can enable the diagnostic network to focus on important parts of the image during processing, saving computational resources and improving the network's processing efficiency. This article integrates the channel attention mechanism and spatial attention mechanism, introduces CBAM [28] (Convolutional Block Attention Module), enhances the network's problem-solving ability, and improves the accuracy of potato leaf disease diagnosis. The CBAM's process for feature maps is shown in Figure 14.

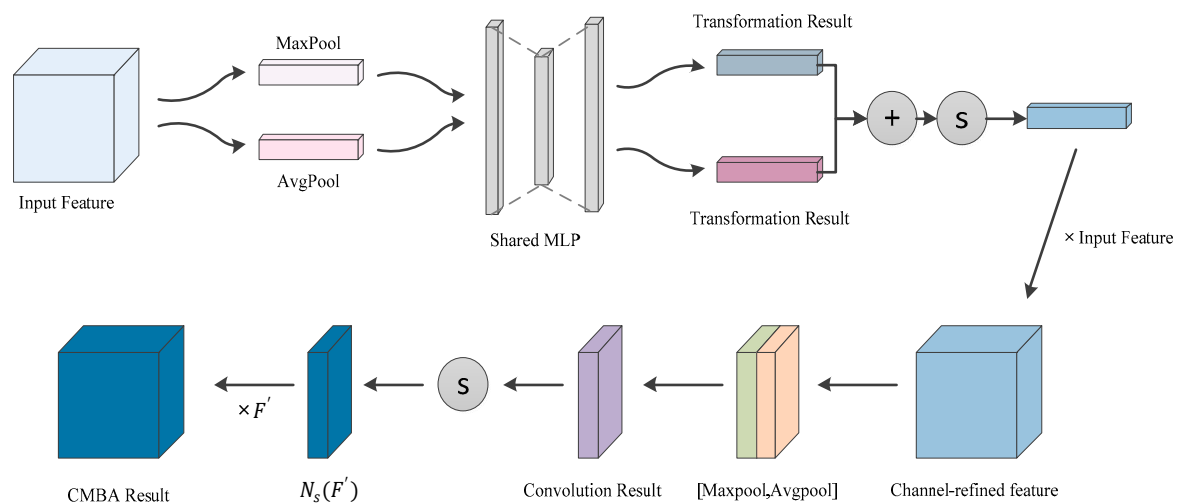


Figure 14. CBAM's process for feature maps.

2.4.3. Activation Function Improvement

The network VGG16 uses the ReLU [29] activation function to enhance its nonlinear ability. However, the ReLU activation function has a one-sided suppression characteristic, which suppresses key neurons and leads to the loss of important classification features, resulting in neuron death and a decrease in network recognition accuracy. The Leaky ReLU activation function inherits the advantages of simple calculation and high efficiency of the ReLU activation function while avoiding the problem of neuron death in the ReLU activation function by introducing slope coefficients. Therefore, this article uses the Leaky ReLU [30] activation function instead of ReLU to enhance the network's ability to extract nonlinear features. The mathematical expression for Leaky ReLU is shown in Equation (4).

$$f(x) = \begin{cases} x, & x > 0 \\ \alpha \cdot x, & x \leq 0 \end{cases} \quad (4)$$

2.5. Hyperparameter Optimization Based on Response Surface Analysis Method

2.5.1. Plackett–Burman Screening Test

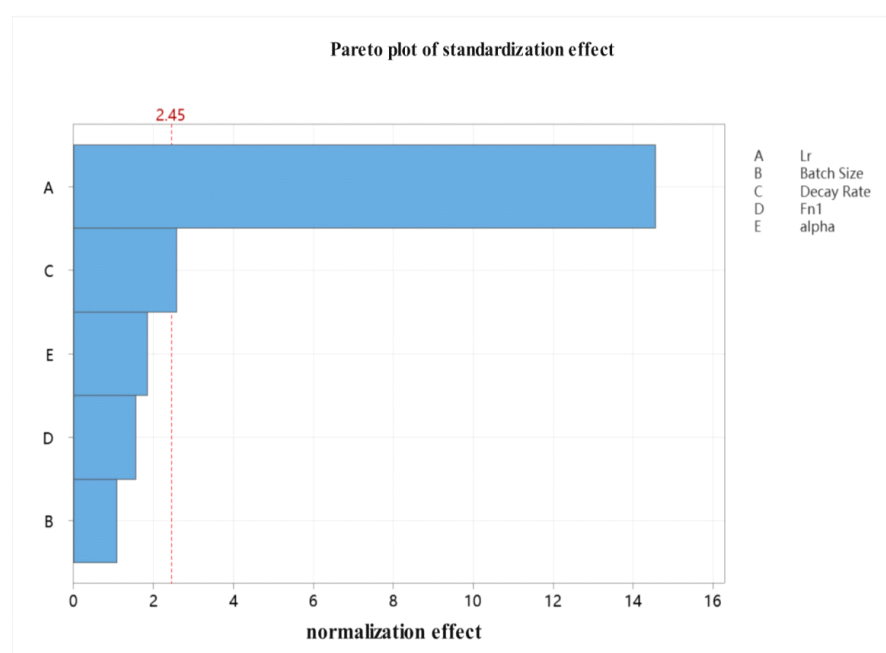
The Plackett–Burman test selected five hyperparameters, including network learning rate (Lr), training set sample size (Batch Size), negative half interval slope of Leaky ReLU activation function (alpha), decay rate of learning rate (Decay Rate), and number of fully connected layer neurons (Fn1), as test factors. Minitab 2021 software was used to perform an analysis of variance on each factor, and the results are shown in Table 3.

Table 3. Plackett–Burman test ANOVA.

	Freedom	Seq SS	Distribution	Adj SS	Adj MS	F-Value	<i>p</i> -Value
Model	5	0.826461	97.41%	0.826461	0.165292	45.11	0.000
linearity	5	0.826461	97.41%	0.826461	0.165292	45.11	0.000
Lr	1	0.776225	91.49%	0.776225	0.776225	211.82	0.000
Batch Size	1	0.004256	0.50%	0.004256	0.004256	1.16	0.323
Decay Rate	1	0.024354	2.87%	0.024354	0.024354	6.65	0.042
Fn1	1	0.008976	1.06%	0.008976	0.008976	2.45	0.169
alpha	1	0.012649	1.49%	0.012649	0.012649	3.45	0.113
error	6	0.021987	2.59%	0.021987	0.003664	-	-
Total	11	0.848448	100.00%	-	-	-	-

where Seq SS is the sequential Sum of Squares, Adj SS is the Adjusted Sum of Squares, and Adj MS is the Mean Square.

According to the ANOVA table, the *p*-values of Lr and Decay Rate are both less than 0.05, indicating a significant impact on the network. According to the Pareto plot of the normalization effect in Figure 15, it can be seen that the contributions of Lr and Decay Rate to the network recognition accuracy are both greater than the significance reference standard line of 2.45, indicating significant contributions. Therefore, it can be concluded that Lr and Decay Rate have a significant impact on the recognition accuracy of the network.

**Figure 15.** Pareto plot of the normalization effect of each hyperparameter.

2.5.2. Steepest Climbing Test Design

A climbing test was conducted on two significant factors, Lr and Decay Rate, in 10 equally spaced intervals. Based on the recognition accuracy of potato leaf diseases in the test set, the optimal range of significance parameters was determined according to the results of the climbing test. The design and results of the steepest climbing test are shown in Table 4.

Table 4. Steepest climbing test design.

Test Group	Lr	Decay Rate	Accuracy
1	0.0001	0.950000000	0.9510
2	0.0002	0.911111111	0.9670
3	0.0003	0.872222222	0.9201
4	0.0004	0.833333333	0.9244
5	0.0005	0.794444444	0.6581
6	0.0006	0.755555556	0.6837
7	0.0007	0.716666667	0.4590
8	0.0008	0.677777778	0.7583
9	0.0009	0.638888889	0.7423
10	0.0010	0.600000000	0.3429

The test results indicate that as Lr increases and the Decay Rate decreases, the accuracy of the test set for the potato leaf disease diagnosis network increases first and then decreases. In group 2 (Lr = 0.0002, Decay Rate = 0.911111111), the recognition accuracy of the network test set reached a maximum of 0.967. Take the parameter values of group 2 as the center point of the response surface analysis test, the parameter values of group 3 as low-level reference, and the parameter values of group 1 as high-level references for the response surface analysis experiment.

2.5.3. Construction and Analysis of Parameter Space Response Surface

Two hyperparameters, Lr and Decay Rate, significantly correlated with the recognition accuracy of potato leaf disease diagnosis network were screened through the Plackett–Burman test, and the optimal test intervals ($L_r \in [0.0001, 0.0003]$, $Decay Rate \in [0.872222222, 0.95]$) for two factors were determined through the steepest climbing test. Based on this, a Central Composite Design (CCD) was designed in the Design Expert 13 software to determine the recognition accuracy of the potato leaf disease diagnosis network and the functional relationship between hyperparameters. A response surface model was constructed, and the optimal hyperparameter combination was predicted. The Central Composite Design and results are shown in Table 5.

Table 5. CCD and test results.

Std	Run	Factor1 A:Lr	Factor2 B:Lr	Response Accuracy (%)
4	1	0.000300000	0.950000	97.02
6	2	0.000341421	0.886100	87.22
1	3	0.000100000	0.822200	84.66
2	4	0.000300000	0.822200	92.65
12	5	0.000200000	0.886100	96.27
3	6	0.000100000	0.950000	95.10
11	7	0.000200000	0.886100	95.10
7	8	0.000200000	0.795732	91.80
5	9	0.0000585786	0.886100	81.68
13	10	0.000200000	0.886100	96.38
10	11	0.000200000	0.886100	91.91
9	12	0.000200000	0.886100	96.06
8	13	0.000200000	0.976468	95.42

The variance analysis conducted in the CCD test is shown in Table 6. The p -value of the model is less than 0.05, indicating that the model is significant and can effectively explain the variability of the data. The p -value of the Lack of Fit is greater than 0.05,

indicating that the model's Lack of Fit is not significant, meaning that the model can fit the data well. The determination coefficient of the overall model $R^2 = 0.8335$, indicating that its regression relationship can explain 83.35% of the variation in the dependent variable. That is, if the independent variable remains unchanged, the degree of variation in the dependent variable will decrease by 83.35%. Adeq Precision = 8.8920, indicating that the response surface model signal is sufficient. The p -values of A-Lr and B-Decay Rate are 0.0488 and 0.0318, respectively, both less than 0.05. It can be concluded that these two test factors still have a significant impact on the recognition accuracy of the potato leaf disease diagnosis network in CCD. Overall, the response surface model established in this test is reliable.

Table 6. Analysis of variance for the CCD test model.

Source	Sum of Squares	DF	Mean Square	F-Value	p -Value	
Model	0.02430000	5	0.00490000	7.0100	0.0119	significant
A-Lr	0.00390000	1	0.00390000	5.6700	0.0488	-
B-Decay Rate	0.00500000	1	0.00500000	7.1600	0.0318	-
AB	0.00090000	1	0.00090000	1.3300	0.2869	-
A ²	0.01420000	1	0.01420000	20.4500	0.0027	-
B ²	0.00000296	1	0.00000296	0.0043	0.9497	-
Residual	0.00490000	7	0.00070000	-	-	-
Lack of Fit	0.00340000	3	0.00110000	3.2500	0.1424	not significant
Pure Error	0.00140000	4	0.00040000	-	-	-
Cor Total	0.02910000	12	-	-	-	-

$R^2 = 0.8335$, Adeq Precision = 8.8920

where DF is freedom.

Using the data processing function in Design Expert 13 software, the fitting function equation of the response surface model can be calculated. After encoding, the equation is shown in Equation (5):

$$\text{Accuracy} = 95.142 + 2.3266284258493A + 2.5961816369738B - 1.2975AB - 4.455375A^2 + 0.11962500000001B^2 \quad (5)$$

where A is Lr and B is Decay Rate.

According to the model fitting equation, the 3D response surface and contour maps were drawn using Design Expert 13 software, as shown in Figures 16 and 17. As shown in Figure 16, with the Decay Rate remaining constant, the recognition accuracy of the potato leaf disease diagnosis network shows a trend of first increasing and then decreasing with the change of Lr, and with the Lr remaining constant, the recognition accuracy increases first then decreases with the increase of Decay Rate. As shown in Figure 17, the contour density along the Lr direction is greater than that along the Decay Rate direction, indicating that the hyperparameter Lr has a greater impact on the recognition accuracy of the potato leaf disease diagnosis network than the Decay Rate.

The results of optimizing the response surface are shown in Figure 18. The red punctuation in the figure represents the optimal factor value position, and the blue punctuation represents the maximum response position. As shown in Figure 17, when the hyperparameters of the potato leaf disease diagnosis network are set to Lr = 0.000207687 and Decay Rate = 0.95, the recognition accuracy of the network is the highest, and the recognition accuracy of the response surface prediction network reaches 97.72%. Following the recommended response surface parameter settings, the network was trained and validated, resulting in a recognition accuracy of 97.87% on the test set. The test observations deviated from the predicted values of the response surface model by

0.15%, achieving the expected results. It can be concluded that the response surface model has good predictive performance, and the obtained fitting equation can describe the relationship between factors and indicators well.

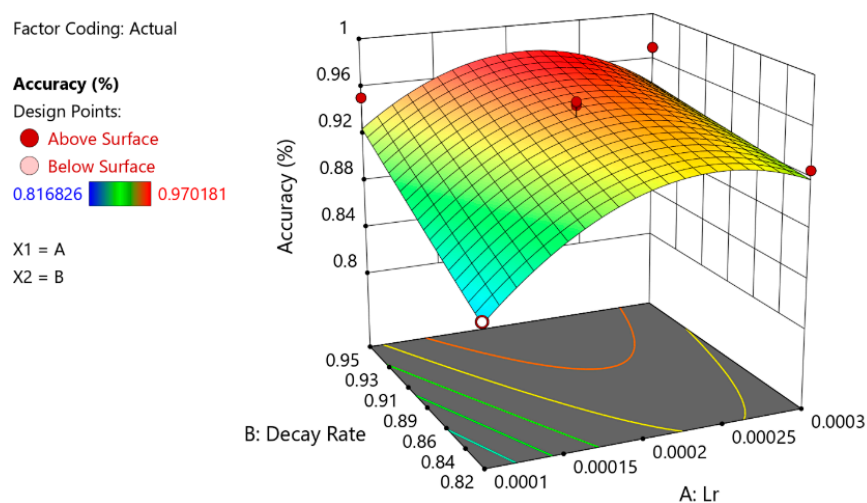


Figure 16. A 3D response surface map.

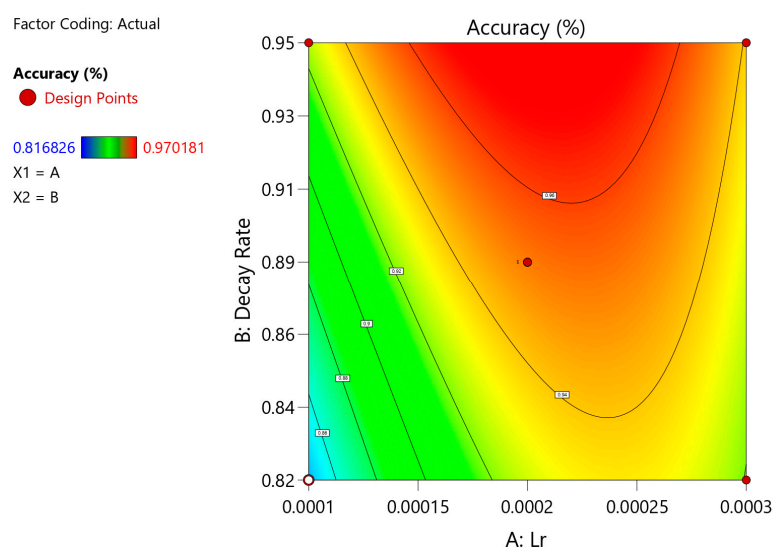


Figure 17. Response surface contour map.

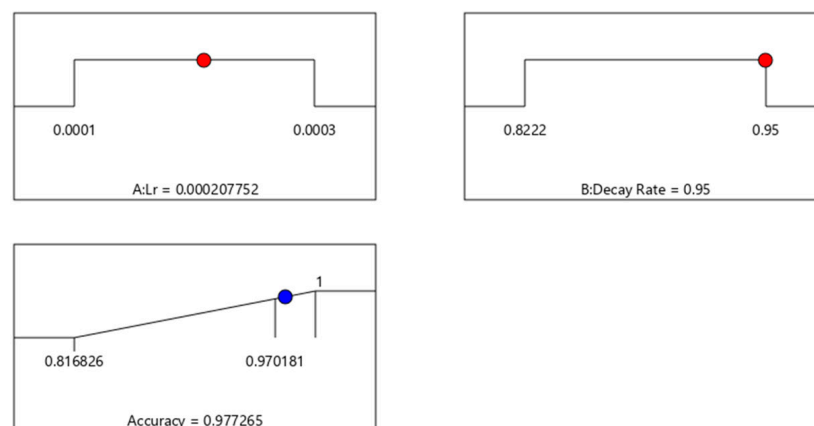


Figure 18. Combination of response surface prediction hyperparameters.

3. Results

3.1. Performance Analysis of Improved Network VGG16S

The improved network structure based on VGG16 is shown in Figure 19. We introduced the CBAM attention mechanism after the first convolution processing module, Block1, in VGG16 to improve the efficiency of the network in processing image information. By replacing the ReLU activation function of the network with the Leaky ReLU activation function, the feature extraction ability of the network is improved while effectively avoiding the gradient vanishing problem caused by neuron death. The network structure of VGG16, which consists of “convolutional layer + flattening layer + fully connected layer”, has been changed to “convolutional layer + global average pooling”, effectively reducing the number of parameters in the model. The improved network can minimize the number of parameters and improve the interpretability of the network while not affecting the recognition accuracy of the network. The improved diagnostic network has fewer parameters and stronger performance compared to VGG16, so it is named VGG16S, meaning small.

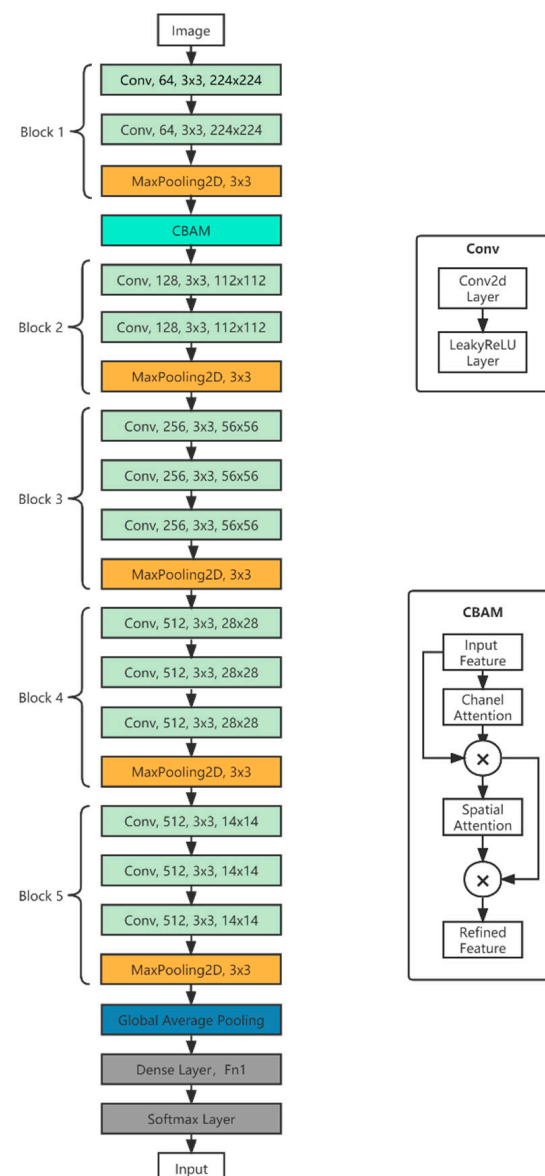


Figure 19. VGG16S network structure diagram.

The VGG16S diagnostic network for potato leaf diseases was trained, and its validation set accuracy is shown in Figure 20.

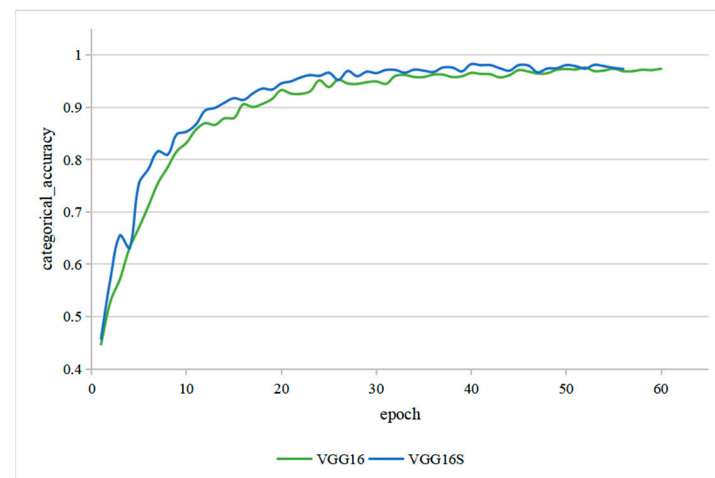


Figure 20. Variation curve of VGG16 and VGG16S network performance with iteration period.

After 25 epochs of training, the convergence curve of the network gradually flattened, and the training ended at the 56th epoch due to the network no longer converging. The final validation set accuracy was 97.34%. The accuracy of the validation set of VGG16S is not significantly improved compared to 97.26% of VGG16, but it can be seen from the training process that the convergence speed of the network VGG16S training is enhanced compared to VGG16. In terms of parameter count, the VGG16S has a parameter count of 14,733,805, which is approximately 15 M. Compared with the 138 M parameter count of the VGG16 network, it reduces 89.36%, greatly reducing the size of the network and making it easier for subsequent deployment and practical use.

The statistics on the number of each network parameter and the recognition accuracy of the testing set are shown in Table 7. The parameter quantities of VGG16, MobilenetV1, Resnet50, Vit, and VGG16S networks are 138 M, 4 M, 25 M, 87 M, and 15 M, respectively. MobilenetV1 has the smallest network volume, followed by the improved network VGG16S proposed in this paper. Although MobilenetV1 has a smaller network size, its lightweight design does not effectively balance the network recognition rate. The recognition accuracy of MobilenetV1's test set is only 59.94%, far lower than VGG16S's 97.87%. VGG16S, while ensuring high recognition accuracy, has a smaller network volume than the other four mainstream image classification networks. Compared to VGG16, the improved network VGG16S only occupies 1/10 of its volume, requiring less storage space and lower deployment costs, meeting the requirements of network lightweighting. As shown in Figure 21, overall, the VGG16S network improved in this paper has the best comprehensive performance.

Table 7. Statistics on the number of network parameters and recognition accuracy of the testing set.

	VGG16	MobilenetV1	Resnet50	Vit	VGG16S
Number of parameters (M)	138	4	25	87	15
Recognition Accuracy (%)	97.23	59.96	81.36	97.02	97.87

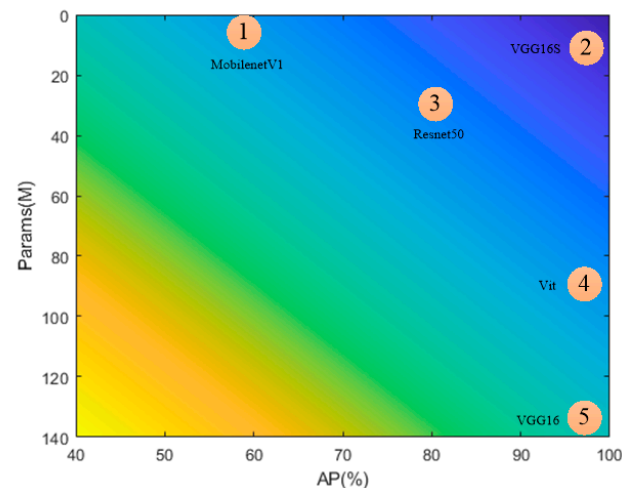


Figure 21. Network performance comparison.

The recognition accuracy of VGG16S for various diseases is shown in Table 8. For normal potato leaves, there were a total of 292 images in the testing set, of which 291 were correctly identified with an accuracy rate of 99.66%; for potato early blight leaves, all 325 images in the testing set were correctly identified with a recognition accuracy of 100%, for potato virus disease leaves, out of a total of 322 images in the testing set, 303 were correctly identified with an accuracy of 94.10%. The VGG16S network has demonstrated high disease recognition capabilities for the three types of potato leaf images, with an accuracy rate of over 94%; the average accuracy rate for the three types of leaf images is 97.87%. This network has strong recognition performance, accurate and reliable recognition results, and can meet the needs of disease diagnosis. The diagnostic effect of the potato leaf disease diagnosis network on input images is shown in Figure 22.

Table 8. Recognition accuracy of VGG16S for different kinds of images.

	Potato Normal Leaves	Potato Early Blight Leaves	Potato Virus Disease Leaves
Number of training sets (pcs)	683	757	750
Number of testing sets (pcs)	292	325	322
Number of correctly identified items in the testing set (pcs)	291	325	303
Recognition Accuracy (%)	99.66	100.00	94.10
Average Accuracy (%)		97.87	

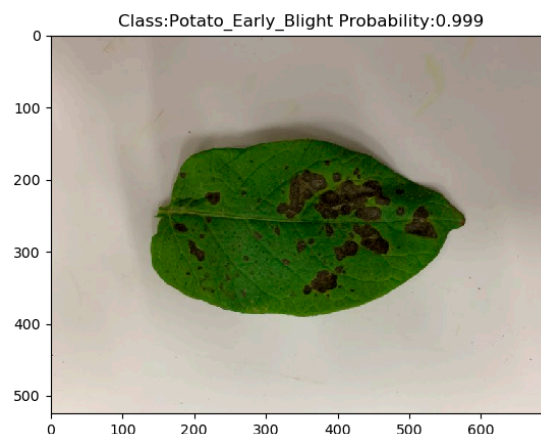


Figure 22. Diagnostic network for potato leaf diseases (VGG16S).

3.2. Hyperparameter Optimization Results and Analysis

Train the network based on the optimal hyperparameters predicted by the response surface methodology above; Figures 23 and 24 show the training process of the potato leaf disease diagnosis network VGG16S after parameter optimization, compared with the unoptimized VGG16S network and four common image classification networks, VGG16, MobilenetV1, Resnet50, and Vit. The training employs a learning strategy with exponential decay of the learning rate and early stopping. This means that the model's training does not specify a specific upper limit on the number of training epochs. The model will automatically stop training when the validation loss value does not decrease over 10 consecutive training epochs.

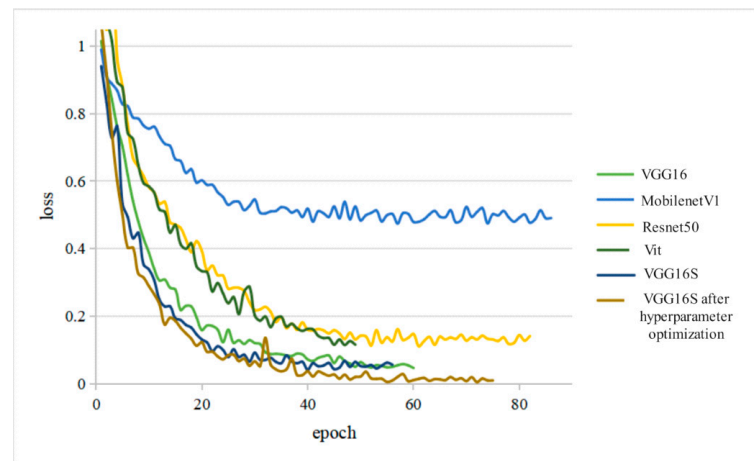


Figure 23. Variation curve of loss with iteration period for each image classification network.

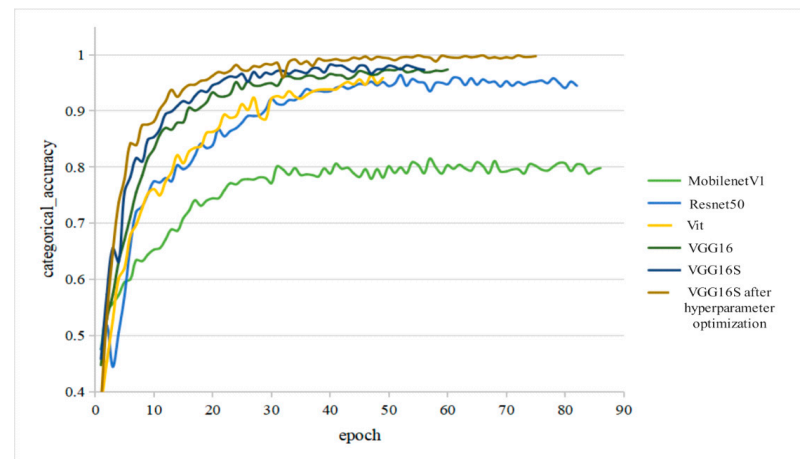


Figure 24. Variation curve of recognition accuracy with iteration period for each image classification network.

As shown in Figure 23, after 30 epochs of iterative training, all six networks began to exhibit convergence characteristics. The convergence rate of the network is VGG16S (after optimization) > VGG16S > VGG16 > Vit > Resnet50 > MobilenetV1. From Figure 24, it can be seen that the validation set accuracy of the network is VGG16S (after optimization) > VGG16S > VGG16 > Resnet50 > Vit > MobilenetV1. Therefore, it can be concluded that the network optimized by hyperparameters in this article outperforms other similar networks in terms of convergence rate during network training and accuracy in the validation set.

3.3. Ablation Study

In order to explore the impact of each improvement strategy on the model performance (average recognition accuracy), a set of ablation experiments was designed in this study. The results of the ablation experiment are shown in Table 9. The model accuracy using only the improved global average pooling strategy, CBAM attention mechanism, or Leaky ReLU activation function is 85.20%, 97.76%, and 97.44%, respectively. It can be seen that using a global average pool instead of the fully connected layer of VGG 16 reduces the total number of parameters in the model while greatly reducing the accuracy of model recognition. The accuracy of adding the CBAM attention mechanism to the global average pooling strategy is 89.46%, and the accuracy of adding the Leaky ReLU activation function to the global average pooling strategy is 88.07%. The accuracy of combining the CBAM attention mechanism and Leaky ReLU activation function is 98.08%, but the large number of model parameters makes training difficult. The accuracy of the improved strategy (the global average pooling + CBAM attention mechanism + Leaky ReLU activation function) is 97.87%.

Table 9. The impact of different improvement strategies on model accuracy.

VGG16	Global Average Pooling	CBAM Attention Mechanism	Leaky ReLU Activation Function	Test Set Identified the Average Accuracy
✓	-	-	-	97.23%
✓	✓	-	-	85.20%
✓	-	✓	-	97.76%
✓	-	-	✓	97.44%
✓	✓	✓	-	89.46%
✓	✓	-	✓	88.07%
✓	-	✓	✓	98.08%
✓	✓	✓	✓	97.87%

3.4. VGG16S Validation Effect on Public Datasets

To validate the performance of the improved model, the plant disease public dataset Plant Village was used for training validation in this study. A total of 2152 potato leaf images were included in the Plant Village dataset, including 152 healthy leaf images, 1000 late blight leaf images, and 1000 early blight leaf images. The 2152 images were divided according to the ratio of the training set and test set of 7:3, and the VGG16S was trained and tested. As shown in Table 10, the identification accuracy of healthy leaves, early disease leaves, and late blight leaves is 95.65%, 95.67%, and 96.00%, respectively, and the average accuracy is 95.82%.

Table 10. VGG16S's recognition accuracy of different categories of images on public datasets.

	Healthy Leaves	Early Disease Leaves	Late Blight Leaves
Total number of dataset images	152	1000	1000
Number of images in the training set	106	700	700
Number of images in the test set	46	300	300
Number of correctly identified by test sets	44	287	288
Identification accuracy	95.65%	95.67%	96.00%
Average accuracy		95.82%	

4. Discussion

4.1. Improvement of VGG16 Network

The network VGG16 performs relatively well in potato leaf disease diagnosis tasks; however, VGG16 adopts a network structure of “convolutional layer + flattening layer + fully

connected layer”, with a network parameter count of 138 M. Therefore, a new, improved algorithm, VGG16S, was proposed to solve the problem of large network parameters.

First, VGG16S adopts the idea of a global average pooling process (GAP) and adopts the structure of “convolutional layer + GAP + fully connected layer”. GAP calculation simplifies the transition connection between the convolutional neural network and the final classification, greatly simplifies the parameter count of the network, and increases the interpretability of the network for image classification.

Second, introduce an attention mechanism. Attention mechanisms in deep learning are mainly divided into channel attention mechanisms and spatial attention mechanisms. The channel attention mechanism can achieve a targeted allocation of network computing power. The spatial attention mechanism can enable diagnostic networks to focus on important parts of the image during processing, saving computational resources and improving network processing efficiency. This article combined the channel attention mechanism and spatial attention mechanism and introduced CBAM (Convolutional Block Attention Module) to improve the network’s problem-solving ability and enhance the accuracy of potato leaf disease diagnosis.

Third, improve the activation function. VGG16 uses the ReLU activation function to enhance its nonlinear ability; there will be no gradient dispersion problem like the Sigmoid and tanh activation functions in their positive and negative saturation regions. However, the ReLU activation function can cause neuronal death, leading to a decrease in network recognition accuracy. VGG16S adopted the Leaky ReLU activation function to avoid neuronal death by introducing a slope coefficient.

The improved VGG16S network has a parameter size of 15 M, and the recognition accuracy of the test set is 97.87%; it not only helps deploy models on resource-constrained devices but also accelerates the inference process of the model and improves real-time performance. Compared with VGG16, the number of network parameters only accounts for 1/10, and the equipment requirements for training the network have also been reduced. The improved network meets the requirements of a lightweight diagnostic network while improving accuracy.

4.2. Perform Hyperparameters Optimization on the VGG16S Diagnostic Network for Potato Leaf Diseases

The problem of improving network performance by adjusting hyperparameters is called Hyperparameter Optimization, which essentially involves finding a set of optimal hyperparameter combinations in a parameter space containing n hyperparameter configuration domains to achieve the minimum loss function or maximum accuracy of the diagnostic network on the dataset. At present, classic hyperparameter optimization search methods include Grid Search (GS), Random Search (RS), Bayesian Optimization (BO), and Genetic Algorithm (GA) [31–33].

This article proposes a new hyperparameter optimization method based on Response Surface Analysis, which mainly consists of the following steps. First, the Plackett–Burman screening experiment was used to screen the five hyperparameters of the network, and two hyperparameters exhibiting high significance with the accuracy of the network testing set were obtained: the learning rate Lr and the learning rate Decay Rate. Second, the steepest climbing test was designed to find the optimal setting interval for hyperparameters, and the CCD center composite response surface test was designed on the interval to construct a response surface model of the relationship between local parameters, and diagnostic network testing set recognition accuracy, the relationship function between hyperparameters and diagnostic network testing set accuracy was fitted to predict the optimal combination of hyperparameters. The results showed that the performance of the disease diagnosis network reached its optimal level when Lr = 0.000207687 and Decay Rate = 0.95.

4.3. Ablation Study and the Test on Public Datasets

To enhance the completeness and credibility of this study, an ablation study was designed in this article. From the results, it can be seen that the individual global average pooling improvement strategy has no positive impact on model accuracy, while the CBAM attention mechanism and Leaky ReLU activation function both have a positive impact on accuracy. In the combination experiment, adding the CBAM attention mechanism separately to the global average pooling strategy can improve the average accuracy of model recognition, which also indicates that the CBAM attention mechanism can improve the recognition accuracy of the model. Similarly, the improvement strategy of the Leaky ReLU activation function can also improve the average accuracy of model recognition, but the effect is weaker than the improvement strategy of the CBAM attention mechanism. In addition, the combination of the CBAM attention mechanism and the Leaky ReLU activation function has the highest accuracy, but the number of model parameters is too large. Therefore, considering the number of model parameters and recognition accuracy, the improvement strategy of global average pool + CBAM attention mechanism + Leaky ReLU activation function is the optimal improvement strategy.

To improve the replicability of the algorithm, the model proposed in this article was tested on the publicly available dataset PlantVillage, and the average accuracy is 95.82% for healthy leaves, early disease leaves, and late disease leaves, with relatively high accuracy.

4.4. Limitations and Development Directions

This article collected images of two common potato diseases (early blight and viral disease) to construct a dataset, while other potato diseases (such as late blight and anthracnose) have not been studied yet. Furthermore, there is no evidence to support the use of datasets for disease images with complex background environments.

In future research, the research team will collect more types of potato leaf disease images to provide a dataset for related research, conduct research on a potato leaf disease diagnosis system in complex environments, establish a potato disease grading model, and accurately apply pesticides to potato diseases according to national standards. Deploy GUI for the diagnosis network of common potato leaf diseases, realize disease image acquisition, local upload, and recognition. Based on the network's diagnosis results of images, provide accurate pesticide application suggestions according to national standards and better guide farmers in scientific potato production practice. This study will reduce the spread of disease spots on tubers, decrease the spread of diseases between plants, and improve the appearance and quality of potatoes.

5. Conclusions

This study focuses on potato early blight and viral diseases. Firstly, images of leaf diseases are collected, and a disease dataset is constructed. The dataset is then expanded using data augmentation methods to increase its quantity and diversity. Secondly, four classic deep learning image classification networks (VGG16, MobilenetV1, Resnet50, and Vit) are used to train the dataset, and VGG16, the classification network with the highest disease recognition accuracy, is selected as the basic research network. A potato leaf common disease diagnosis network VGG16s based on a deep learning algorithm is obtained by lightweight improvement of the basic network, the fusion of attention mechanism, and introduction of the activation function. Finally, to further improve the performance of the diagnostic network, this study used response surface analysis to construct the local hyperparameter spatial contour of the network for hyperparameter optimization and hyperparameter configuration optimization. This article also completed ablation experiments and public dataset testing. The test results show that the algorithm proposed

in this paper improves the diagnostic performance of the network and has a certain degree of generality.

This article establishes a diagnostic method for potato leaf diseases, and this study provides a theoretical basis for scientifically effective disease prevention and control, reducing pesticide misuse and abuse, and achieving green prevention and control.

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